Performance Testing Specification

Snowflake Data Warehouse & Tableau Enterprise Analytics

Version: 1.0
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Status: Approved

Executive Summary: This technical document defines the load testing methodology, target key performance indicators (KPIs), execution strategies, and benchmark criteria for validating the high-concurrency performance of the Snowflake Data Warehouse engine and Tableau Enterprise reporting platform.

1. SYSTEM ARCHITECTURE & LOAD SCOPE

WORKLOAD ISOLATION & PIPELINE TARGET ARCHITECTURE

Snowflake Engine (Data Layer)

- Multi-Cluster Virtual Warehouses
- Auto-Scaling & Result Caching Validation
- Heavy Aggregation & ELT Stress Tests

Tableau Server (Analytics Layer)

- VizQL Server & Data Engine Capacity
- Extract Hyper Engine vs. Live Pushdown
- Concurrent Browser Interactive Sessions

2. SNOWFLAKE PERFORMANCE BENCHMARK SCENARIOS

ID	TEST SCENARIO	WAREHOUSE	LOAD	TARGET P95	SLA CRITERIA
SF-01	Point Lookup / Single Record	X-Small	50 Users	< 800 ms	99.5% Success Rate
SF-02	Aggregation & Joins (Medium Data)	Medium	20 Users	< 3.0 s	Zero Remote Spillover
SF-03	Large ELT Transformation	Large	5 Jobs	< 45.0 s	Memory Budget < 85%
SF-04	Auto-Scaling Stress Test	Med (1-5 C)	500 Users	< 5.0 s	Scale-up < 30 sec

Snowflake Execution Note: Result caching (`USE_CACHED_RESULT = FALSE`) must be disabled during primary baseline execution runs to measure raw compute capacity.

3. TABLEAU PERFORMANCE BENCHMARK SCENARIOS

ID	TEST SCENARIO	TYPE	SESSIONS	TARGET P90	PASS CRITERIA
TB-01	Executive Dashboard Initial Render	Extract (Hyper)	100 Sessions	< 2.5 s	VizQL CPU < 80%
TB-02	Operational Detail Dashboard	Live Pushdown	50 Sessions	< 4.0 s	Query Pushdown < 3s
TB-03	Interactive Action Filtering	Extract	75 Sessions	< 1.5 s	Zero Gateway Timeout
TB-04	Parallel Extract Refresh Schedules	Backgrounder	10 Jobs	< 15.0 min	Zero Failure / Retry

4. KEY METRIC TARGETS (KPI SUMMARY)

≤ 3.0s
P95 QUERY RESPONSE

≤ 85%
MAX VIZQL CPU LOAD

< 0.1%
TARGET ERROR RATE

5. RISK ANALYSIS & OPTIMIZATION GUIDELINES

Snowflake Disk Spilling: If high-concurrency queries trigger spilling to remote cloud storage, adjust the warehouse scaling policy to *Economy* mode or optimize table micro-partition clustering keys prior to re-testing.

Tableau VizQL Bottlenecks: For dashboards exceeding the 2.5-second initial render SLA, convert complex inline calculated fields to materialized columns inside Snowflake.

6. PROJECT SCHEDULE & SIGN-OFF MATRIX

PHASE / MILESTONE	OWNER	PLANNED DURATION	STATUS
Test Scripting & Environment Setup	QA / Performance Team	Days 1 - 3	Completed
Snowflake Baseline & Load Execution	Data Engineering	Days 4 - 6	In Progress
Tableau Load & VizQL Stress Testing	BI Engineering Team	Days 7 - 8	Scheduled
Final Technical Sign-off & Reporting	Enterprise Architect	Day 9	Pending